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Turn Charts for Interaction Analysis: Visually Mapping the Conversation Floor

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Abstract: In the Learning Sciences and CSCL communities, interaction analysis (IA) is an important methodological approach to studying teaching and learning. IA relies on transcripts to selectively encode, represent, and experience video. However, researchers using IA continue to highlight the need for more dynamic and interactive transcription techniques to interpret, experience, and even speculate about data in context. In this paper, we use a comparative case study to introduce a transcription technique we call *turn charts.* Turn charts integrate ideas in IA with interactive and alternative visualization to visually map conversation over time, while also providing dynamic ways to select and experience audio and video data. Findings highlight how turn charts extend current transcription techniques to (1) answer common questions in IA, (2) advance exploratory modeling in the early stages of IA, and (3) bridge IA and visualization research to expand how people, things, and context are represented.

Introduction

In the Learning Sciences and CSCL communities, interaction analysis (IA) is an important methodological approach to studying teaching and learning (Jordan & Henderson, 1995; also see Hall & Stevens, 2015). For example, scholars from diverse theoretical orientations continue to use and expand IA to characterize embodied activity, unpack complex and technologically mediated interactions, conceptualize affect, and develop alternative understandings of interaction that foreground relational ontologies and histories across diverse settings including classrooms, museums, cities, and the natural world (see DeLiema, Enyedy & Danish, 2017; Davidsen & Ryberg, 2017; Ehret & Hollett, 2014; Hennessy Elliott, 2020; Keifert & Stevens, 2019; Kelton, 2021; Krishnamoorthy et al., 2021; Ma & Munter. 2014; Marin et al., 2020; Philip et al., 2018; Silvis et al., 2022; Solomon et al., 2022; Steier et al., 2019; Taylor, 2017; Vogelstein et al., 2019; Vossoughi et al., 2020).

At its core, IA relies on the creation of transcripts to selectively encode, represent, and experience audio and video data. Such transcripts support interpretive and other kinds of analyses of learning or teaching, often from sociocultural, social practice, and critical perspectives. However, IA researchers continue to highlight the need for more dynamic and interactive transcription techniques to interpret, experience, and even speculate about data in context (Mathur & Shapiro, 2022; Silvis, 2019; Vossoughi et al., 2020).

In this paper, we use a comparative case study approach to introduce a new transcription technique we call *turn charts.* Turn charts integrate ideas in IA with interactive and alternative visualization techniques to map conversation and multimodal interaction over time, while also providing dynamic ways to experience audio and video data. Findings and discussion highlight how this technique extends current transcription techniques to (1) answer common questions faced in IA, (2) advance exploratory modeling in the early stages of IA, and (3) build new connections between IA and visualization research to expand how people, things, and context are represented for qualitative research in the Learning Sciences and CSCL communities. We conclude by emphasizing several limitations of this work and next steps to extend this work in ways that further expand the methodological commitments and possibilities of IA (see Krishnamoorthy et al., 2021; Marin et al., 2020; Silvis et al., 2022).

Theoretical framework

Transcription and interaction analysis

Transcription is a process used by qualitative researchers to encode and translate unstructured data such as audio or video into a representation that supports interpretive analysis. Creating a transcript is often characterized as a form of modeling, a selective process that is either explicitly or implicitly directed by theory (Ochs, 1979; also see Derry et al., 2010; Hall, 2000). Thus, any transcription technique encodes and represents data in ways that reflect the assumptions, biases, and goals of the researcher doing the transcribing.

How transcripts are structured and what aspects of talk-in-interaction are included (or excluded) are quite variable in IA. Some researchers focus on the content of what is said by people, paying less attention to how conversations are sequentially organized. Others use fine-grained orthographic standards such as those developed

by Gail Jefferson and used in conversation analysis (Jefferson, 1984) to model the structure of talking (e.g., prosody and breath), turn exchange, and self or other repair in fine-grained detail. More recently, researchers have expanded beyond a focus on transcribing language to include other communicative modalities (e.g., gesture, gaze, or bodily configurations; see Bezemer & Mavers, 2011; Kress & van Leeuwen, 2006). For example, researchers in Learning Sciences or CSCL often integrate video still frames and text in transcripts to model and analyze embodied activity in diverse settings (DeLiema, Enyedy & Danish, 2017; Goodwin, 2017; Keifert & Stevens, 2019; Ma & Munter, 2014; Kelton, 2021; Marin et al., 2020; Silvis et al. 2022; Steier, 2014; 2022; Solomon et al., 2022; Vogelstein et al., 2019). Such transcripts are useful not only to interpret embodied activity but to develop more expansive views of learning that often foreground equity and social justice perspectives.

While no single transcription system can serve all purposes, two themes in this body of work motivate our efforts. First, apart from a few novel examples (see Angus et al., 2012; Erickson, 2004), transcripts are typically produced with language-based tools such as Microsoft Word and PowerPoint that support static and typically text-focused transcripts. This is the case even as transcription has rapidly expanded in innovative ways from a focus only on language to multimodality, with multimodal transcripts often using the same tools to characterize phenomena such as gesture. Second, little work has explored how to link transcripts with video to augment exploratory modeling and experiences in the early stages of IA in ways that can shape subsequent analysis and findings. We seek to develop interactive visualizations for IA to support visual exploration of and experiences with transcripts earlier in the process of IA.

Interactive and alternative visualization

We also draw from recent developments in the field of information visualization, building on recent work calling for new methods that bridge visualization and qualitative research to interactively explore relations between people, things, and contexts.

Historically, the field of information visualization has focused on the exploration, analysis, and presentation of data. Classic visualization frameworks or pipelines describe visualization as a visual mapping process, which involves transforming data into data tables and visual structures that facilitate the exploration and analysis of data (Card, Mackinlay, and Schneiderman, 1999). These frameworks have placed a strong emphasis on the visual design of representations, including the use of different representational forms and color.

However, in the past decade, there have been two shifts in the visualization community that inform our work. First, researchers have come to recognize that interacting with representations is just as crucial as visual design when it comes to exploring, analyzing, or presenting data (see Yi et al., 2007). Interaction in visualization can take various forms, including zooming, panning, filtering, highlighting, and linking. Second, new theories of visualization have also emerged that focus on alternative ways of knowing, non-binary approaches to data, and the contextual dimensions of data (e.g., D'Ignazio & Klein, 2016; Lupi, 2017). These theories respond to broader arguments that visualization must more meaningfully and intimately represent and communicate with people and communities. For example, ideas such as Data Humanism explore ways data visualization can be more human and experiential through interaction techniques, playful animations, glyphs, and a focus on small data as opposed to big data (Lupi, 2017).

Turn charts draw from this work to focus on developing interactive visualizations that explore new ways to interact and animate transcripts and video data about people, things, and context in ways that embody ideas like Data Humanism. Likewise, they are designed to address common questions or problems in IA in ways that align with recent efforts to develop more animated and experiential transcription techniques for IA (see Hall & Stevens, 2015; Shapiro & Silvis, 2023; Shapiro et al., 2024; Vossoughi et al., 2020).

Method

Method and data sources

Our subsequent analysis uses a comparative case study approach to explore two video recordings collected in classroom and museum contexts. The first recording is a well-known and influential two-minute video of classroom interaction that shows the teaching of a key idea in the physics of matter—that matter occupies space—in a kindergarten-first grade bilingual classroom. In the video, a teacher is seated with her students around her on the floor of a classroom and conducts a brief experiment involving a jar and sand to illustrate how matter occupies space. The teacher also facilitates a discussion about the experiment with the class. This dataset is a publicly available dataset originally collected by Kris Gutiérrez and colleagues (see Gutiérrez et al., 1999; Gutiérrez, 2008; https://youtu.be/d8_pRUR-hmg) and has influenced Learning Sciences scholars in a myriad of ways (see https://www.learninghowtolookandlisten.com). The second dataset is multi-perspective video of a five-member family's approximately 9 minute visit to a gallery space in a cultural heritage museum. This data set was collected

through our prior work as part of a larger three-year project to understand how visitors learn about the diverse historical and cultural heritage of American Roots and Country music. As part of this project, we collected multi-perspective video of 22 families' complete museum visits in collaboration with museum partners.

We selected these video recordings for several reasons. First, they represent different types of video data commonly collected and studied by IA researchers, allowing us to demonstrate the applicability of turn charts to diverse video data. Second, the datasets vary in terms of the scale of collaborative interaction, the ways they were recorded, and the settings in which they were recorded. As a result, we can compare turn charts at different scales and demonstrate how differences in data collection and video recording influence the resulting visualizations. Third, each of these datasets is publicly available, making it possible to share interactive visualization tools from this work widely. Finally, each dataset is well known to educational researchers through prior publications and analyses, allowing us to compare how turn charts expand previous perspectives on each video dataset.

Positioning participant-researcher-data-methods

We are mindful of the many ways that video-based research with people subjects them to the interpretive gaze of persons (researchers, readers of research reports) they do not know and will never meet. Conducting IA with research participants or taking seriously the advice that IA can bring together the perspectives of diverse analysts may mitigate these issues somewhat, but video-based research unavoidably risks centering some experiences, while excluding or erasing others. In our own analytical practices, we attend with care to how talk and action related to learning are demonstrably relevant to persons in the video recording, and we try to avoid interpretive stances that objectify talk and action as reflections of unobservable beliefs or values held by persons in the video recording. We also acknowledge that in analyzing the first case, people in the recording are not known to us, and although these recordings are available for public use, each has traveled far from original settings or unfolding situations for meaningful human activity. We also acknowledge the many ways in which unequal relations of power exist between researchers and participants, adults and children, and between human activity recorded as "data" and the lived experiences of persons in that activity. While turn charts by no means dismantle these relations of power, we believe they can be used in ways that call attention to aspects of human activity that might otherwise be overlooked or erased (e.g., conversational parity, how participation changes and how topics unfold over phases of recorded activity).

Case 1 analysis

Turn charts are built on the concept or metaphor of a conversation floor. They help to see the density and coverage of a speaker's talk, treating the elapsed time of the recording as a visual ground for mapping participation over time. Particularly, they provide new ways to see who is speaking, at what times, and when overlapping talk occurs (i.e., a contest over getting the floor). At the same time, they also support new ways of reading transcripts and exploring and experiencing video data. In the following, we first describe the visual encoding and interaction techniques of turn charts and then describe several insights and experiences made visible through visual exploration with turn charts of this dataset.

Visual encoding

Figure 1 shows a turn chart of the focal data set. In the figure, each ellipse represents a conversation turn spoken by either the teacher or a student from the focal transcript. Color indicates the speaker, and ellipses are organized across a conventional timeline or time series display from 0 (left) to 120 seconds (right). The height of the ellipse indicates the number of words in each conversation turn. Thus, taller ellipses indicate more words spoken during a conversation turn. The width of an ellipse represents the duration of a turn at talk, with possibilities for overlapping talk at turn boundaries. Thus, wider ellipses indicate longer conversation turns and where ellipses overlap indicates overlapping speech. Importantly, turn charts are not static representations, but rather, support several ways to interact and experience transcripts and video data detailed in the following section.

Figure 1
Screenshot from Interactive Turn Chart
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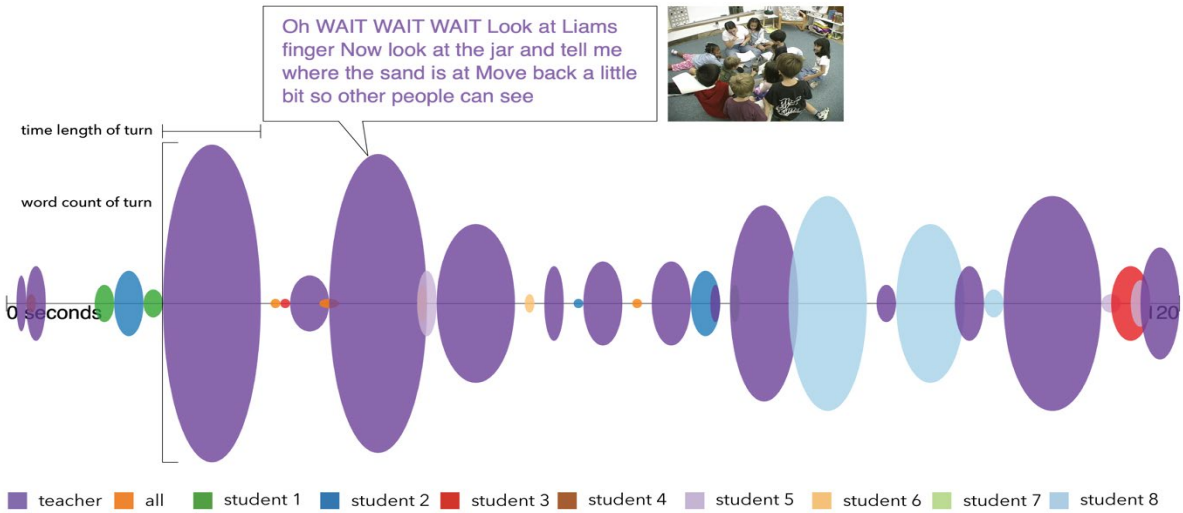


Each ellipse is a conversation turn where color encodes the speaker. Ellipses are organized across a timeline from left to right where the height of the ellipse indicates the number of words in each conversation turn, and the left and right of each ellipse indicates the beginning and end of each conversation turn. Users can hover over ellipses to read conversation turns and click on an ellipse or the timeline to play video of a turn at talk at that moment in the recording

Interaction techniques

As Figure 1 begins to convey, turn charts support several different ways to interact with and animate data. Namely, the figure shows how hovering over an ellipse allows one to display and read each conversation turn. The Turn Chart also provides ways to interact with video (if loaded) by clicking anywhere on the timeline, which begins playing video from that moment in time. In this way, one can quickly interact with and explore transcript and audio/video data in new ways.

Figure 2 further summarizes several different kinds of interaction techniques. First, one can use a turn chart to select different speakers or actors. For example, Figure 2A and 2B display turns from the teacher and all students respectively. As a result, one can quickly compare contributions from different speakers or isolate contributions to support more detailed exploratory analysis. Second, Figure 2C illustrates how one can also change the visual layout of turn charts to separate speakers vertically. Such a display draws inspiration from musical scoring techniques used by interaction analysts such as Erickson (2004) as well as recent efforts to convey choreographies of interaction that draw inspiration from dance (Solomon et al., 2022; Vogelstein et al., 2019). This layout allows different views and comparisons across all speakers, in this case. Finally, Figure 2D illustrates how one can use an interactive timeline to zoom in on sequences of interaction—in this case we have zoomed in on the first 10 seconds of this interaction and the timeline and sizing of the circles adjust accordingly. This allows one to explore relations between parts and wholes in transcribed data early in the process of IA (Erickson, 2006).

As before, across all these different views, one can hover to read conversation or click to watch and experience audio and video data. Moreover, one can also "animate" the transcript at different speeds, including in real time. Doing so plays the transcript, and circles grow in size and appear as conversation turns are spoken providing a different way to interact and experience data.

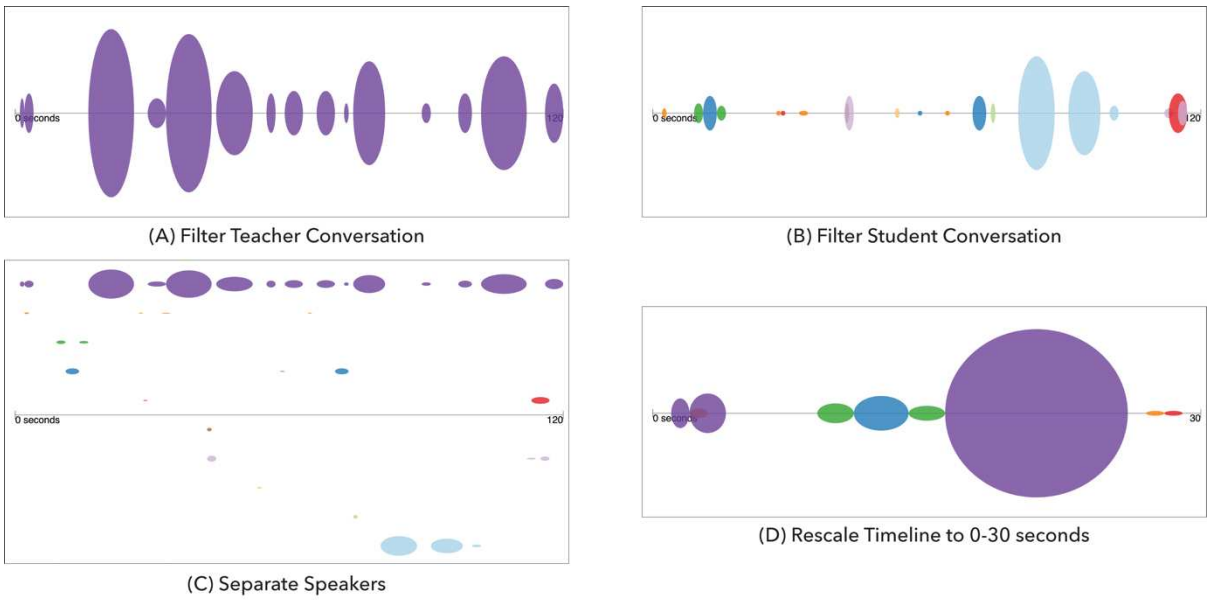
Insights and experiences

For the focal transcript and video, several insights and experiences are made visible that expand previous understandings of this dataset. First, one can quickly see, compare, and reflect upon the distribution of turns of talk over time. For example, the turn chart and particularly Figure 2B clearly makes the appearance and continued conversation of Student 8 towards the end of the recording highly visible, suggesting that she plays a substantive role from that point forward in the activity. Likewise, the teacher speaks for much longer stretches of time and makes much longer conversation turns in comparison to her students. Notably, nearly all students contribute to this interaction while the amount and timing of their contributions can be quickly studied and compared.

Second, one can observe and characterize the pacing of the conversation (i.e., length of conversation turns with respect to both the number of words and amount of time). For example, the pace of conversation between the teacher and students is slower (e.g., there are periods of time when no one speaks) as the interaction

starts, which in this case reflects the teacher setting up the science experiment that grounds this interaction. Moreover, moments when students speak for longer periods, often while asking a question, or rapid exchanges where students speak one after the other are quickly identifiable.

Figure 2
Illustration of Different Interaction Techniques Available with Turn Chart



(A) filtering only the teacher's conversation turns, (B) filtering only students' conversation turns, (C) changing the layout to display speakers separated vertically, and (D) rescaling the timeline to zoom in on the first 10 seconds of the interaction.

Third, overlapping ellipses provide a way to see overlapping talk (when start and stop times are included in the transcript), both at turn boundaries and, as a type of interaction, over time. In this case, we can see how students speak over one another, but they do not speak over or interrupt the teacher's speech. More generally, the turn chart provides ways to identify, explore and experience moments of potential interest from transcripts of talk and to conduct closer analysis of talk in interaction with video. For example, it is straightforward to visually explore video from each of the teacher's or student's individual conversation turns or moments of silence.

Case 2 analysis

While space constraints prevent a complete examination of our second case, our second case analysis elaborates on our previous analysis to illustrate different dimensions of turn charts at a larger scale. Figure 3 shows a turn chart of a five-member family's approximately nine-minute visit to a museum gallery space. Color once again indicates the speaker. As before, we illustrate how one can play video by clicking along the timeline. Specifically, we have played video from a conversation turn spoken by a 10 year old boy named Jeans at the start of an engagement with an exhibit of a famous musician named Bill Monroe.

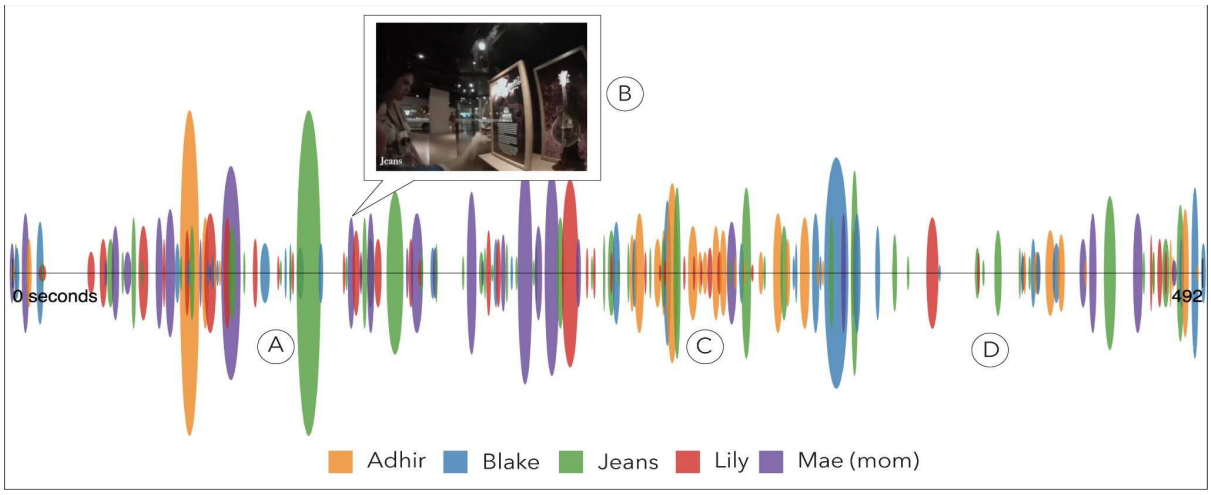
Like before, this turn chart reveals several phenomena important to IA that are challenging to see with conventional transcripts. First, it vividly represents the overall conversation distribution of the family during their visit to this gallery space. Each family member's contributions, both in volume and frequency of talk, are evident. Second, conversational dynamics such as rhythm, intensity, and overall pace are emphasized. A notable pattern marked in the figure is the rise and fall of conversation as the family engages with and departs from exhibits (Figure 3B). Such patterns, which previous studies characterize as "engagement contours" (Shapiro et al., 2017), are shown here from a different perspective. Third, the turn chart illustrates periods of rapid turn exchanges and overlapping conversation turns, as exemplified by the exchange between Adhir and his fiancé Lily (Figure 3C).

Furthermore, the turn chart highlights several potentially significant moments for closer analysis. Extended conversation turns, as well as brief exchanges, are highlighted. Likewise, moments of silence are made equally visible, underscoring ways to characterize and explore speech and silence and more generally, variable qualities of sound in teaching and learning contexts (see Reimers, 2022).

Finally, the turn chart also underscores one significant limitation: Family members may not always be in close proximity, leading to separate but concurrent conversations occurring in different parts of the gallery

space. Put differently, the lack of spatial encoding leads to potential challenges of turn charts for learning spaces where movement plays an essential learning role. While turn charts could be expanded to address such challenges, such challenges also underscore the need to develop techniques that emphasize space and movement.

Figure 3
Interactive Turn Chart of a Five Member Family's Visit to a Museum Gallery Space



Color indicates the speaker. (A) one potentially significant conversation turn from Jeans, (B) video is played from the beginning of an engagement contour between family members at the Bill Monroe Exhibit, (C) an engagement contour focused on remembering the life and death of Hank Williams between Adhir and Lily and (D) slowing of conversation as the family begins to shift to the next gallery space.

Discussion

In summary, turn charts offer IA researchers new ways to explore and make sense of who holds the conversation floor as collaborative interaction unfolds. They also provide alternative ways to animate, interact with, select, and experience transcript and video data to reveal new phenomena about teaching and learning. While we have shown their value for two cases, they can be used for a variety of audio and video datasets of different sizes. We suggest they have several implications for IA researchers, along with limitations that can be expanded upon in future work to support different kinds of projects, including participatory research with practitioners such as teachers.

First, our analysis highlights how they support answering certain kinds of questions common in IA that are challenging to answer through traditional transcripts. Particularly, turn charts are ideal for analyzing issues of conversational parity—who speaks, when and how much? These kinds of questions regularly arise in IA, and they can be challenging to answer through conventional transcripts. They also offer a visual form and provide precise and dynamically adjustable ways to explore how identified speakers jointly manage the conversational floor over time, allowing one to zoom across different temporal scales and speaker selections.

Second, as we stated in the introduction, most transcripts used in IA are static representations, often created during the later stages of analysis to communicate findings. Turn charts advance transcription techniques to support exploratory modeling in the early stages of IA. For example, we illustrated how they help to identify significant moments and phenomena of interest that can support subsequent analysis. Likewise, we illustrated how they support integrative analysis to notice and index across scales or what qualitative researchers often characterize as "global" and more "molecular" phenomena available in an audio or video recording (Erickson, 2004). Finally, we note that turn charts are useful to make different aspects of the underlying transcript data or model visible and identify potential errors with a transcript model. In other words, turn charts can reveal problems with or theoretical decisions made about how a transcript is encoded (e.g., identifying only one or two speakers in a multi-party activity, inconsistent timing for conversation turns). All of these things are quite useful in the early stages of IA.

Third and more generally, this work contributes to ongoing efforts to develop more animated and interactive visualizations for IA, bridging traditions of IA with interactive visualization techniques (see Mathur & Shapiro, 2022). While our primary focus in this paper has been visually exploring conversation, we continue to expand turn charts to encompass a range of modalities. For example, currently we are developing ways to layer codes for gesture, incorporate material dimensions and more-than-human entities, and build on other ideas drawing inspiration from scholars such as Erickson (2004), Silvis (2019), and Vossoughi et al., (2020) to

characterize spatial and affective dimensions. We are also exploring the use of this technique to support reflective professional practice for teachers and other practitioners to extend traditional video-based reflection.

We plan to make turn charts available through a free and open-source visualization platform that will also share several other kinds of qualitative transcription and visualization techniques in the future. This platform will allow researchers and practitioners to load video and/or traditional transcripts formatted as text files, CSV files, or MS word documents to immediately produce several dynamic visualizations such as turn charts of transcript data. We are currently gathering feedback from the IA community as well as practitioners to enhance and scale this platform in ways that best support IA researchers and practitioners.

While we suggest several contributions of turn charts for IA, we conclude by highlighting several limitations and next steps. First, like all visualizations, turn charts are highly reductive, do not characterize many aspects of everyday life and experience, and are not neutral with respect to dimensions of hierarchy and power (see Buckley, 2013). Second, while we discussed how turn charts could in the future incorporate non-verbal features, our analysis here focused on conversation. Third, our case analysis focused on shorter timescales — for example, issues of inclusion, exclusion, and engagement in classrooms accrue over longer time periods, including relational histories that teachers have with students and communities, and the broader sociohistorical contexts of schools. Such limitations present opportunities to extend this work but also underscore the need for other alternative methods to support IA. Accordingly, we suggest there is a need and opportunity to continue to expand such techniques, asking how interactive transcription techniques can more deeply represent people, things, and notions of context across different time and spatial scales to support traditions of IA as used in the Learning Sciences and CSCL communities.

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